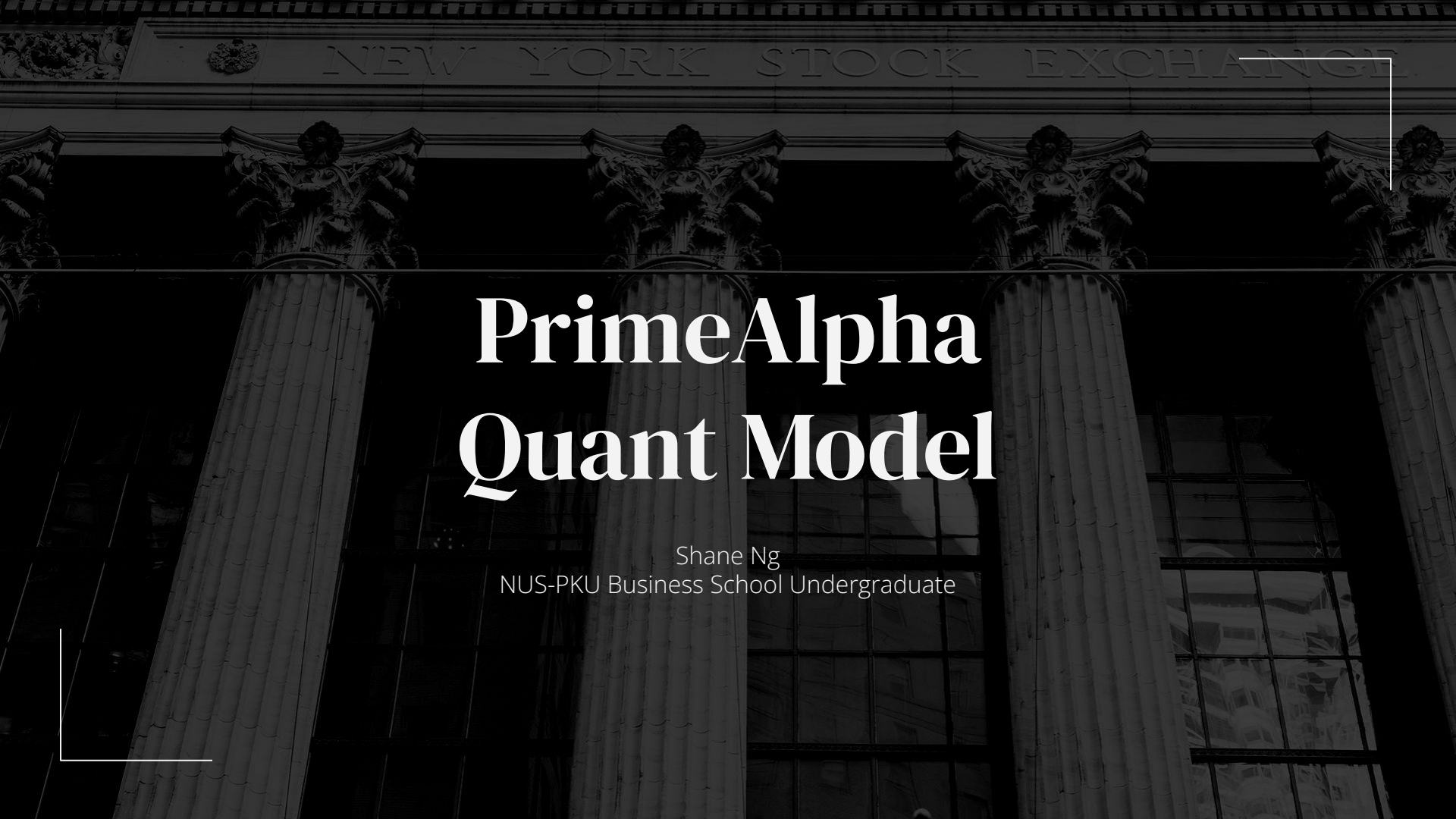




PrimeAlpha Quant Model

Shane Ng
NUS-PKU Business School Undergraduate

Market Leadership Tends to Persist

Winners continue
outperforming.

Drawdowns destroy
compounding.

Regime awareness improves
survivability.

SPY Return
(1998 - 2026)
1,102%

Strategy Return
(1998 - 2026)
14,124%

Momentum Persistence remains one of the strongest documented market anomalies.

Systematic Investment Framework



01

Regime Filter

Goes Risk-on and Risk-off depending on where the SPY is relative to its 200DMA



02

Weighted Scoring

Uses out-of-sample testing to optimise weights for Relative Strength and Trend Momentum



03

Universe Selection

Looks at the S&P 500 and aims to beat the index



04

Portfolio Construction

Equal-weights the top 20 highest scoring stocks



05

Risk Management

Immediately drop stocks that experiences a severe, immediate drawdown

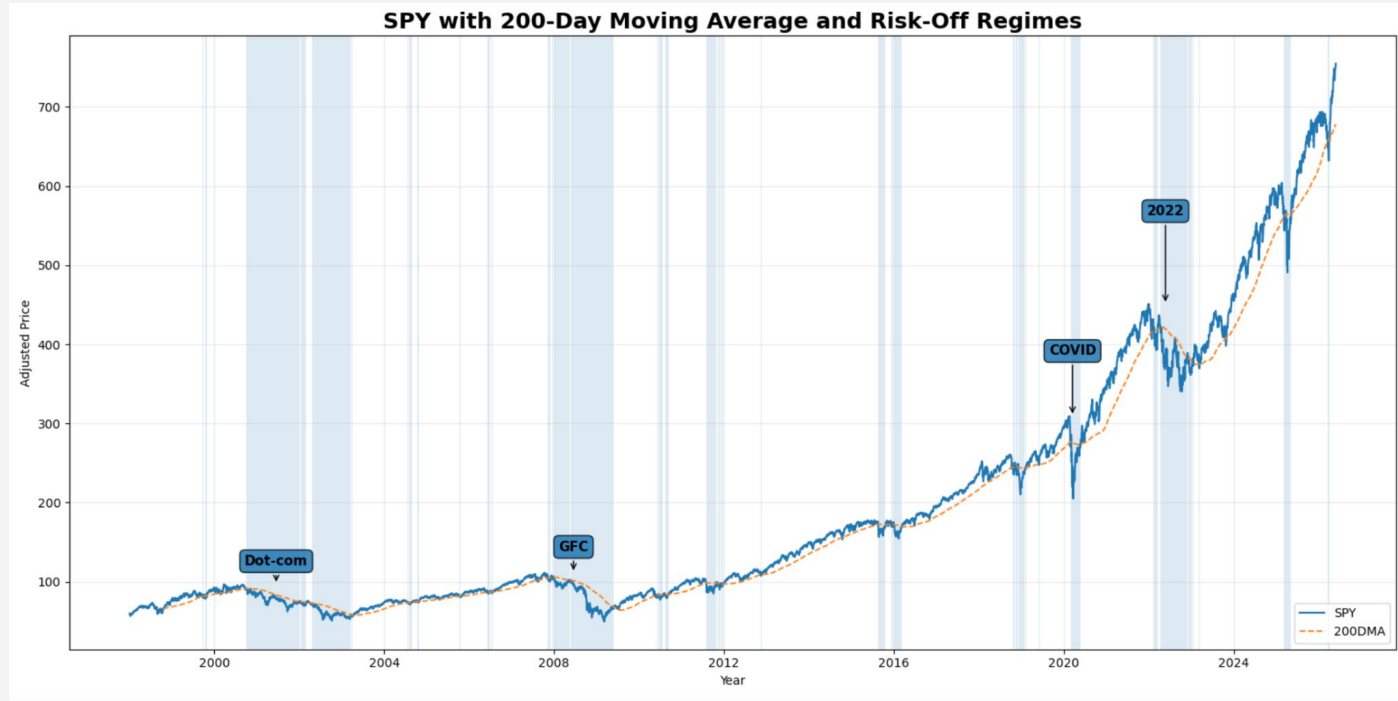


06

Monthly Rebalancing

Compounds returns and lock in profits

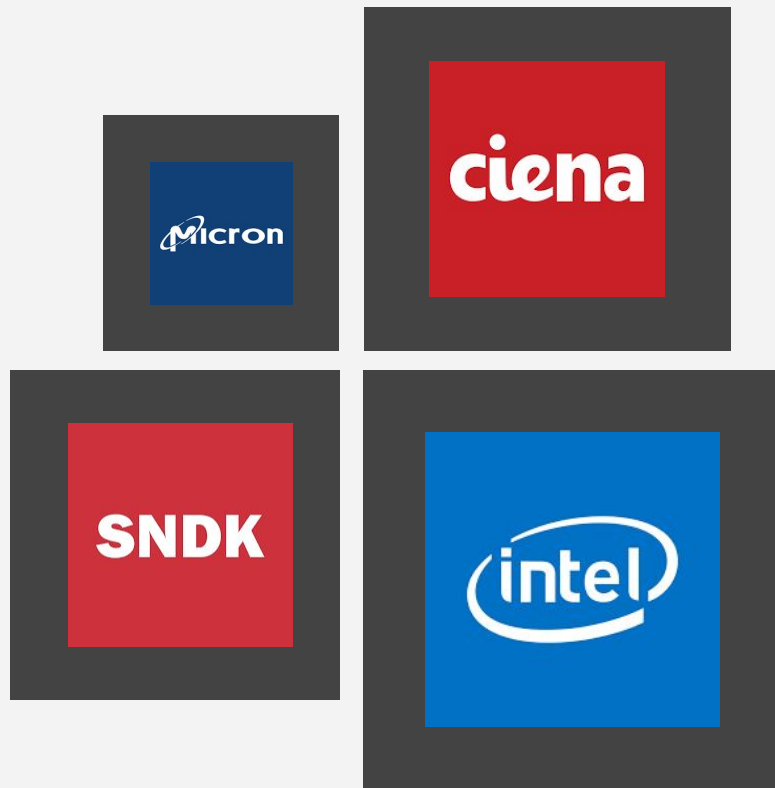
Risk Exposure Adjusts with Market Conditions



Risk-on (White Area): Full Equity Exposure

Risk-off (Blue Area) : Partial Defensive Allocation to Gold/Cash

Capital Rotates Toward Market Leadership



Relative Strength Against the SPY

Scores based on relative outperformance against the SPY

Relative Strength Against the Sector

Scores based on relative outperformance against its sector

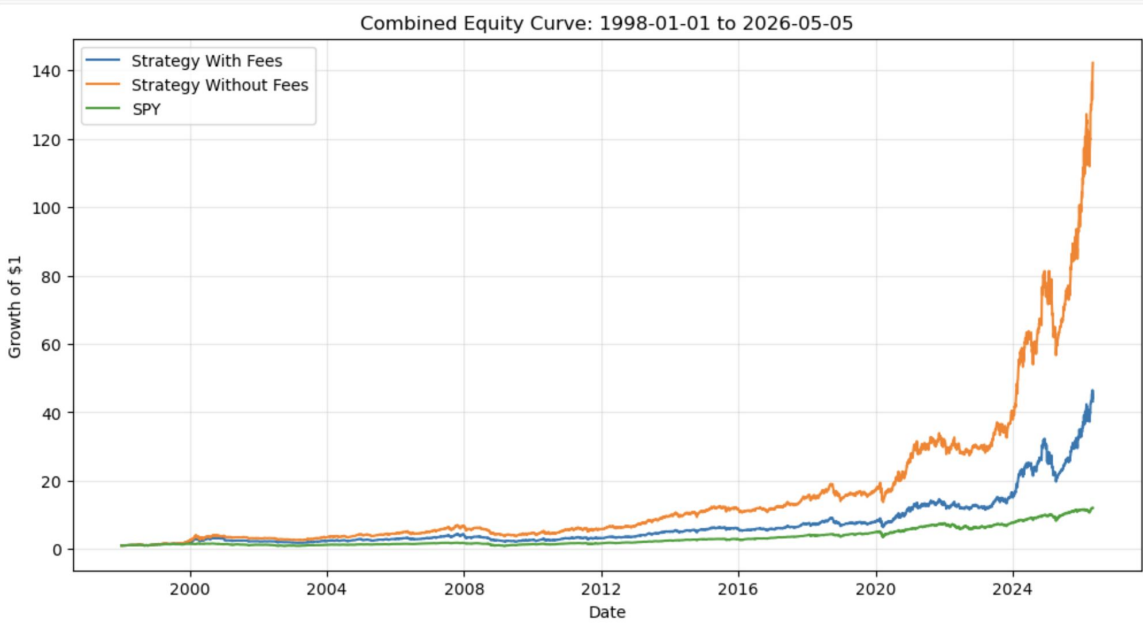
Trend Momentum

Scores based on whether the MA200 is Rising, and whether the 50MA>150MA>200MA

Portfolio Construction



Strong Long-Term Risk-Adjusted Performance

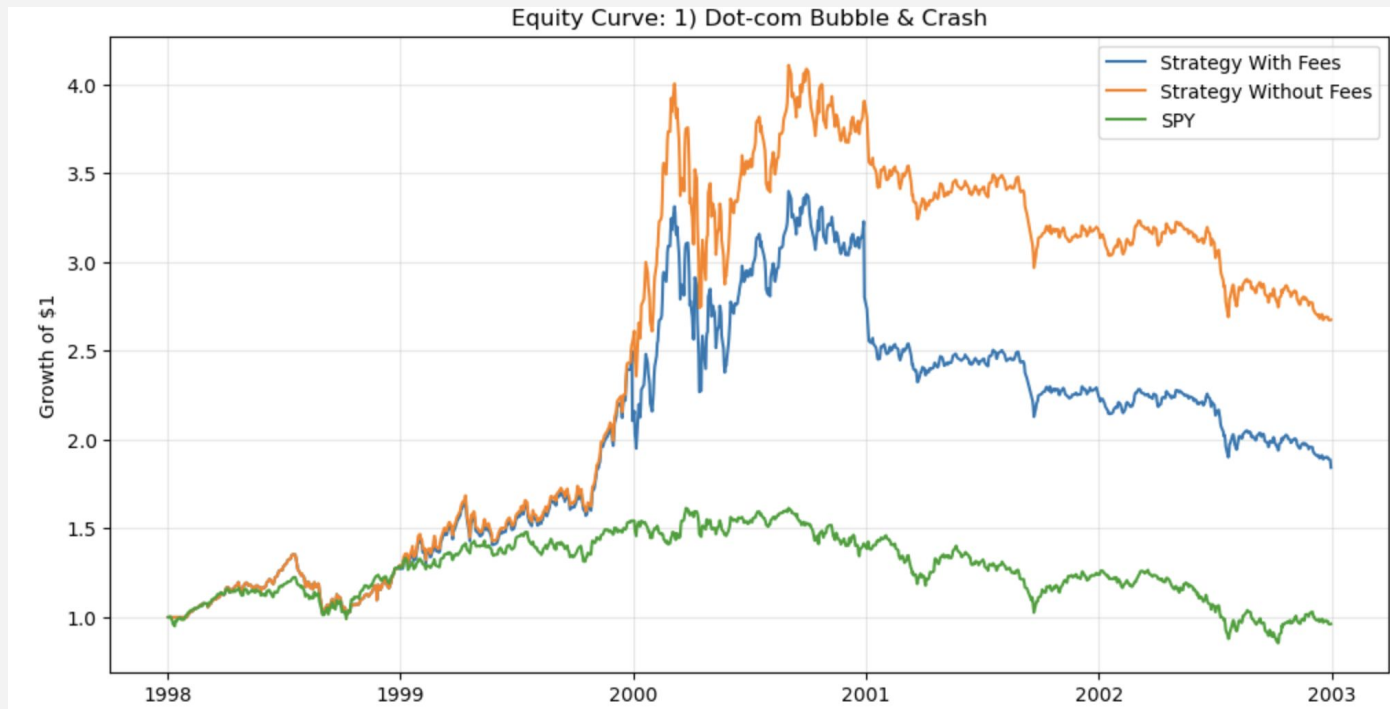


	Metric	Strategy Without Fees	SPY
0	Total Return	14124.22%	1102.70%
1	CAGR	19.16%	9.19%
2	Volatility	19.36%	15.46%
3	Sharpe rf=0	0.99	0.59
4	Max Drawdown	-46.86%	-55.16%

	Metric	Strategy With Fees	SPY
0	Total Return	4219.77%	1102.70%
1	CAGR	14.24%	9.19%
2	Volatility	20.18%	15.46%
3	Sharpe rf=0	0.71	0.59
4	Max Drawdown	-51.60%	-55.16%

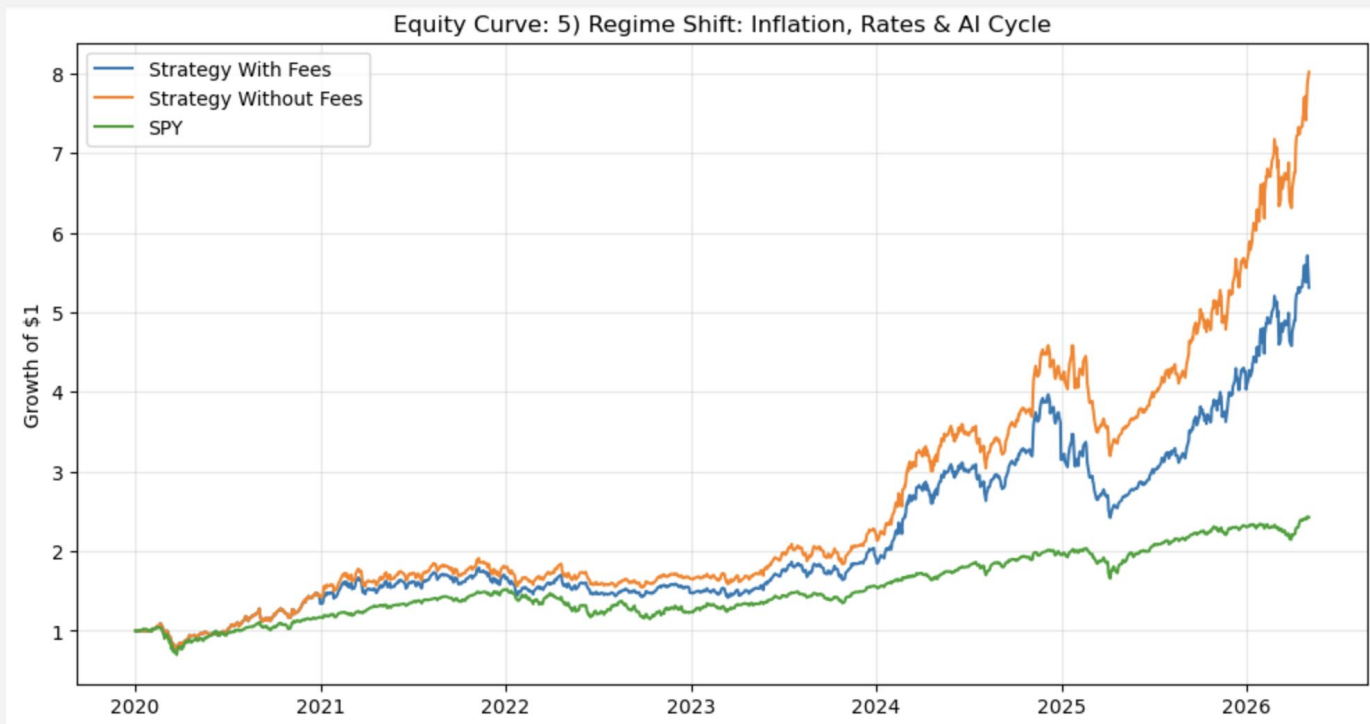
Backtest conducted with out-of-sample testing different from data set used to train the model to prevent overfitting.
SPY components is point-in-time accurate to prevent survivorship bias.

Crisis Resilience in Bear Markets Improves Compounding



Strategy quickly rotates towards quality, defensive equities that leads the market out of a crisis, improving returns.

Market Momentum in Bull Markets drive Alpha



Strategy quickly rotates towards equities that sees strong relative strength and trend momentum, driving returns.

Model Recommendations as of 4 May 2026

	Ticker	Sector	Price	AvgDollarVol50	Ret_3M	Ret_6M	Ret_9M	Ret_12M	SPY_3M	SPY_6M	SPY_9M	SPY_12M
0	CIEN	Information Technology	539.26	1.19B	108.84%	185.98%	503.23%	665.28%	4.12%	6.05%	16.21%	30.01%
1	INTC	Information Technology	97.48	6.58B	106.44%	140.31%	402.34%	384.98%	4.12%	6.05%	16.21%	30.01%
2	LITE	Information Technology	981.53	4.89B	136.82%	382.68%	832.48%	1491.00%	4.12%	6.05%	16.21%	30.01%
3	MU	Information Technology	568.52	18.04B	33.83%	152.82%	441.28%	625.18%	4.12%	6.05%	16.21%	30.01%
4	SNDK	Information Technology	1239.02	13.45B	97.60%	538.93%	2867.36%	3708.27%	4.12%	6.05%	16.21%	30.01%
5	STX	Information Technology	740.35	1.90B	77.54%	182.44%	384.10%	724.01%	4.12%	6.05%	16.21%	30.01%
6	WDC	Information Technology	436.54	2.69B	69.95%	216.87%	474.82%	889.12%	4.12%	6.05%	16.21%	30.01%
7	FIX	Industrials	1879.72	646.93M	62.20%	92.25%	173.62%	355.12%	4.12%	6.05%	16.21%	30.01%
8	VRT	Industrials	330.27	2.19B	77.29%	70.52%	137.08%	259.07%	4.12%	6.05%	16.21%	30.01%
9	COHR	Information Technology	337.42	2.12B	56.49%	152.59%	230.33%	392.33%	4.12%	6.05%	16.21%	30.01%
10	SATS	Communication Services	121.46	753.58M	6.73%	66.22%	330.86%	432.84%	4.12%	6.05%	16.21%	30.01%
11	GLW	Information Technology	161.56	1.84B	51.55%	81.38%	162.92%	266.67%	4.12%	6.05%	16.21%	30.01%
12	ON	Information Technology	103.47	655.24M	70.59%	102.81%	83.94%	159.21%	4.12%	6.05%	16.21%	30.01%
13	ALB	Materials	193.41	378.20M	16.30%	97.55%	190.30%	226.13%	4.12%	6.05%	16.21%	30.01%
14	KEYS	Information Technology	351.54	470.57M	60.09%	103.31%	119.89%	140.91%	4.12%	6.05%	16.21%	30.01%
15	LRCX	Information Technology	257.63	2.37B	9.57%	59.79%	173.82%	260.67%	4.12%	6.05%	16.21%	30.01%
16	AMD	Information Technology	350.93	9.19B	45.60%	36.35%	105.30%	258.23%	4.12%	6.05%	16.21%	30.01%
17	TER	Information Technology	340.93	1.13B	39.05%	95.25%	227.68%	359.51%	4.12%	6.05%	16.21%	30.01%
18	APA	Energy	41.31	328.60M	62.26%	87.55%	126.11%	171.83%	4.12%	6.05%	16.21%	30.01%
19	PWR	Industrials	753.47	650.58M	58.59%	68.59%	91.43%	134.90%	4.12%	6.05%	16.21%	30.01%

Risks and Limitations



Momentum Crashes

Will continue to brainstorm and incorporate risk management guardrails



Lagging Filters

Will use ML Optimization to further research on potential leading indicator candidates



Fees and Transaction Costs

Can be reduced as AUM scales





Thanks

Shane Ng
NUS-PKU Business School Undergraduate